

CS PMOS Circuit Analysis

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Abstract— This report presents the design and simulation of a CS PMOS amplifier using the GF180MCU PDK. Analytical expressions are derived, transistor parameters are extracted from LTspice, and the circuit is designed to meet the required specifications. The calculated results are then compared with simulations.

I. SYMBOLIC ANALYSIS

A. Reference Current I_{REF}

The reference branch contains resistor R, transistor MN3, and transistor MN1. Since they are connected in series, the same current flows through all elements:

$$I_{REF} = I_{D3} = I_{D1}$$

Applying KVL from the supply to ground:

$$V_{SUPPLY} = I_{REF}R + V_{DS3} + V_{DS1}$$

Since transistors MN3 and MN1 are diode-connected, the drain-source voltage equals the gate-source voltage:

$$V_{DS} = V_{GS}$$

Substituting this condition into the KVL equation gives:

$$V_{SUPPLY} = I_{REF}R + V_{GS3} + V_{GS1}$$

Thus, the reference current can be expressed as:

$$I_{REF} = \frac{V_{SUPPLY} - V_{GS3} - V_{GS1}}{R}$$

Also, using the MOS saturation current equation:

$$I_{REF} = \left(\frac{1}{2}\right) k_{n1} (V_{GS1} - V_{TN})^2$$

B. Bias Current I_{BIAS1}

The upper branch of the amplifier contains the PMOS transistors MP1 and MP2. Since they are connected in series, the same current flows through both transistors.

$$I_{BIAS1} = I_{D1} = I_{D2}$$

For a PMOS transistor operating in saturation, the drain current is given by:

$$I_D = \left(\frac{1}{2}\right) k_p (V_{SG} - |V_{TP}|)^2$$

For transistor MP1, the source is connected to the supply voltage. Therefore:

$$V_{SG1} = V_{SUPPLY} - V_{IN}$$

Substituting into the current equation gives:

$$I_{BIAS1} = \left(\frac{1}{2}\right) k_{p1} (V_{SUPPLY} - V_{IN} - |V_{TP}|)^2$$

C. Bias Current I_{BIAS2}

The branch containing MN4 and MN2 forms a cascoded NMOS current mirror. Therefore, the current I_{BIAS2} is a scaled version of the reference current.

The mirror ratio is:

$$m = \frac{\left(\frac{W}{L}\right)_4}{\left(\frac{W}{L}\right)_3} = \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

The bias current becomes:

$$I_{BIAS2} = m I_{REF}$$

D. Voltage Gain v_{out}/v_{in}

The circuit operates as a common-source PMOS amplifier with cascode loading.

The small-signal current produced by transistor MP1 is

$$i_o = g_{m1} v_{sg1}$$

Since the source of the PMOS is AC ground:

$$\frac{v_{out}}{v_{in}} = -g_{m1} R_{out,tot}$$

Where: $R_{out,tot} = R_{up} // R_{down} // R_L$

The resistance looking upward into the PMOS cascode branch is approximately:

$$R_{up} \approx r_{o1} + r_{o2} + g_{m2} r_{o1} r_{o2}$$

The resistance looking downward into the NMOS cascoded current sink is:

$$R_{down} \approx r_{o4} + r_{oMN2} + g_{m4} r_{o4} r_{oMN2}$$

Thus the gain becomes:

$$\frac{v_{out}}{v_{in}} = -g_{m1} (R_{up} // R_{down} // R_L)$$

E. Input Resistance

Since MOS transistor gates draw no current: $i_{in} \approx 0$
Therefore, the input resistance is approximately: $r_{in} \approx \infty$

F. Output Resistance

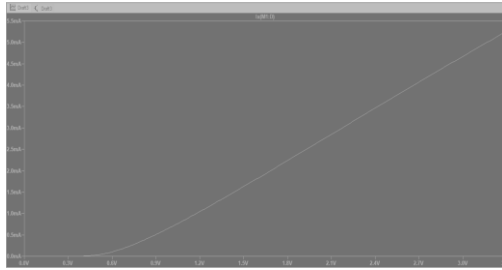
The output resistance is the parallel combination of the resistance looking upward and downward from the output node: $r_{out} = R_{up} // R_{down}$

Substituting the expressions for each resistance:
 $r_{out} = (r_{o1} + r_{o2} + g_{m2}r_{o1}r_{o2}) // (r_{o4} + r_{oMN2} + g_{m4}r_{o4}r_{oMN2})$

II. TRANSISTOR PARAMETER EXTRACTION

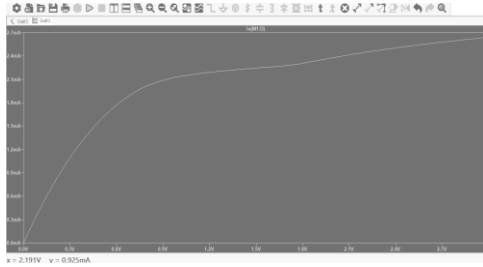
A. NMOS Parameters

1. Threshold Voltage



$V_{th} = 0.6V$

2. Lambda λ



$\lambda = \text{slope}/I_{d0} = 0.0792552647542V^{-1}$

3. Parameter k

Using the MOS saturation equation:

$$k_n = \frac{I_D}{(V_{GS} - V_{th})^2} \approx 1.35 \times 10^{-3} \frac{A}{V^2}$$

4. Parameter k'n

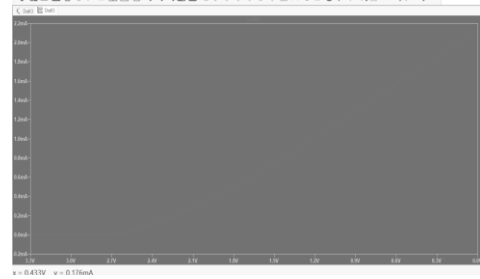
$$k'_n = 2k_n \left(\frac{W}{L} \right)$$

$$\frac{W}{L} = \frac{10}{0.28} = 35.714$$

$$k'_n \approx 0.0756 \frac{mA}{V^2}$$

B. PMOS Parameters

1. Threshold Voltage

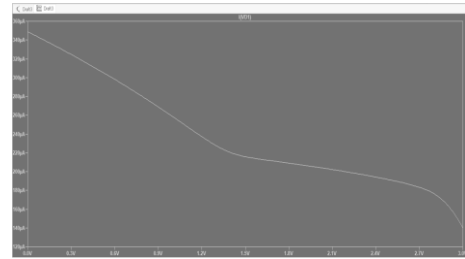


$V_S = 3.3V \quad V_G \approx 2.7V$

$V_{SG} = V_S - V_G = 0.6V$

$|V_{thp}| \approx 0.6V \quad V_{thp} \approx -0.6V$

2. Lambda λ



$$\lambda_p = \left| \frac{\text{slope}}{I_D} \right| \approx 0.113V^{-1}$$

3. Parameter k

$$|I_D| = k_p (V_{SG} - |V_{thp}|)^2$$

$V_{SG} = V_S - V_G = 2.3V$

$V_{SG} - |V_{thp}| = 2.3 - 0.6 = 1.7V$

$$k_p = \frac{|I_D|}{(V_{SG} - |V_{thp}|)^2} \approx 8.94 \times 10^{-5} \frac{A}{V^2}$$

4. Parameter k'p

$$k_p = \left(\frac{1}{2} \right) k'_p \left(\frac{W}{L} \right)$$

$W = 10 \mu m \quad L = 0.28 \mu m$

$$k'_p = 2k_p \left(\frac{W}{L} \right) k'_p \approx 5.0 \times 10^{-6} \frac{A}{V^2}$$

III. HAND CALCULATIONS FOR CIRCUIT DESIGN

The objective is to determine the bias currents and design parameters required to satisfy the following specifications:

$$I_{BIAS2} = \frac{I_{REF}}{5}$$

$$V_{OUT} = 1.65V$$

$$90 < |Av| < 100$$

$$R_L = 30k\Omega$$

A. Load Current

The current flowing through the load resistor is

$$I_L = \frac{V_{OUT}}{R_L} = \frac{1.65}{30\,000} = 55\mu A$$

B. Choice of I_{BIAS1}

To meet the required gain range, the small-signal gain of the amplifier can be approximated as

$$Av \approx -g_m R_L$$

Choosing a target gain near the center of the range,

$$|Av| \approx 95$$

$$\text{Thus } g_m = \frac{|Av|}{R_L} = \frac{95}{30\,000} \approx 3.17mS$$

Since the transconductance of the transistor depends on the drain current, a reasonable design value is selected:

$$I_{BIAS1} = 100\mu A$$

This current will be used as the bias current of the amplifying transistor.

C. Determination of I_{BIAS2}

Applying KCL at the output node,

$$I_{BIAS1} = I_{BIAS2} + I_L$$

Therefore,

$$I_{BIAS2} = I_{BIAS1} - I_L = 45\mu A$$

D. Reference Current I_{REF}

The current mirror relationship is given as

$$I_{BIAS2} = \frac{I_{REF}}{5}$$

Thus,

$$I_{REF} = 5I_{BIAS2} = 225\mu A$$

E. Transistor Width of MP1

For transistor MP1 operating in saturation,

$$I_D = \left(\frac{1}{2}\right) k'_p \left(\frac{W}{L}\right) V_{OV1}^2$$

Rearranging to solve for the aspect ratio:

$$\frac{W}{L} = \frac{(2 I_D)}{(k'_p V_{OV1}^2)}$$

From the previous section: $I_D = I_{BIAS1} = 100\mu A$

The overdrive voltage is obtained from the transconductance relation: $g_m = \frac{2I_D}{V_{OV}}$

$$\text{Thus: } V_{OV1} = \frac{2I_D}{g_m} \approx 0.063V$$

Using the extracted PMOS parameter:

$$k'_p = 5 \frac{\mu A}{V^2}$$

Substituting:

$$\frac{W}{L} = \frac{(2 \times 100\mu A)}{\left(5 \frac{\mu A}{V^2} \times (0.063)^2\right)} \approx 10075$$

Given that $L = 0.28\mu m$

The width becomes $W_{MP1} \approx 2820\mu m$

F. Transistor Width of MP2

For transistor MP2, a reasonable overdrive voltage is selected as: $V_{OV2} = 0.20V$

Since MP2 operates in saturation, the drain current is given by:

$$I_D = \left(\frac{1}{2}\right) k'_p \left(\frac{W}{L}\right) V_{OV2}^2$$

Because MP2 is in series with MP1 in the upper branch, both transistors carry the same bias current. Therefore:

$$I_D = I_{BIAS1} = 100\mu A$$

Solving for the aspect ratio:

$$\frac{W}{L} = \frac{(2 I_D)}{(k'_p \cdot V_{OV2}^2)}$$

Substituting:

$$\frac{W}{L} = \frac{(2 \times 100\mu A)}{\left(5 \frac{\mu A}{V^2} \times (0.20)^2\right)} = 1000$$

Given that $L = 0.28\mu m$

Then $W_{MP2} = 280\mu m$

G. NMOS Widths

The transistors MN1 and MN3 carry the reference current:

$$I_D = I_{REF} = 225\mu A$$

Substituting:

$$\frac{W}{L} = \frac{(2 \times 225\mu A)}{\left(75.6 \frac{\mu A}{V^2} \times (0.2)^2\right)} \approx 148.8$$

Since $L = 0.28\mu m$

Then:

$$W_{MN1} = W_{MN3} \approx 42\mu m$$

For the transistors MN2 and MN4, the current mirror relation is:

$$I_{BIAS2} = \frac{I_{REF}}{5}$$

Therefore, the transistor width scales with the same ratio:

$$W_{MN2} = W_{MN4} = \frac{W_{MN1}}{5} \approx 8.4\mu m$$

H. Reference Resistor R_{REF}

Applying KVL in the reference branch:

$$V_{DD} = I_{REF} \cdot R_{REF} + V_{GS1} + V_{GS3}$$

For NMOS devices:

$$V_{GS} = V_{th} + V_{OV}$$

Using the design values:

$$\begin{aligned} V_{th} &= 0.6V \\ V_{OV} &= 0.2V \end{aligned}$$

Thus:

$$\begin{aligned} V_{GS1} &= 0.6 + 0.2 = 0.8V \\ V_{GS3} &= 0.8V \end{aligned}$$

Substituting:

$$I_{REF} \cdot R_{REF} = 1.7$$

Since $I_{REF} = 225\mu A$

Then:

$$R_{REF} \approx 7.56k\Omega$$

I. Bias Voltage V_{BIAS2}

For the PMOS cascode transistor: $|V_{SG2}| = |V_{thp}| + V_{OV2}$

Using the chosen design values: $|V_{thp}| = 0.6V$ and $V_{OV2} = 0.2V$

$$\text{Thus: } |V_{SG2}| = 0.6 + 0.2 = 0.8V$$

To keep both PMOS devices in saturation, an internal node voltage is selected: $V_X \approx 2.45V$

$$\text{Therefore: } V_{BIAS2} = V_X - |V_{SG2}| \approx 1.65V$$

J. Input Bias Voltage V_{IN}

For MP1: $|V_{SG1}| = |V_{thp}| + V_{OV1}$

Using: $|V_{thp}| = 0.6V$ and $V_{OV1} = 0.063V$

$$\text{Thus: } |V_{SG1}| = 0.663V$$

Since: $V_{SG1} = V_S - V_G$ and $V_S = V_{DD} = 3.3V$

$$\text{Then: } V_{IN} = V_{DD} - |V_{SG1}| \approx 2.64V$$

IV. SMALL-SIGNAL ANALYSIS

A. Voltage Gain ($\frac{v_{out}}{v_{in}}$)

The small-signal voltage gain of the CS PMOS amplifier can be approximated by: $A_v \approx -g_{m1} \cdot ((R_{up} \parallel R_{down}) \parallel R_L)$

First, the transconductance of MP1 is $g_{m1} \approx 3.17 \text{ mS}$

Next, the small-signal output resistance of a MOS transistor is: $r_o = \frac{1}{(\lambda I_D)}$

For MP1 and MP2, the drain current is $100 \mu\text{A}$, so:

$$r_{o,p} = \frac{1}{(0.113 \times 100 \mu\text{A})} \approx 88.5 \text{ k}\Omega$$

For MN2 and MN4, the drain current is $45 \mu\text{A}$, so:

$$r_{o,n} = \frac{1}{(0.079 \times 45 \mu\text{A})} \approx 281 \text{ k}\Omega$$

The upward small-signal resistance of the PMOS cascode branch is approximated as:

$$R_{up} \approx r_{o,MP1} + r_{o,MP2} + g_{m,MP2} \cdot r_{o,MP1} \cdot r_{o,MP2}$$

The transconductance of MP2 is $g_{m,MP2} = 1.0 \text{ mS}$

Therefore:

$$R_{up} \approx 88.5 \text{ k}\Omega + 88.5 \text{ k}\Omega + (0.001)(88.5 \text{ k}\Omega)(88.5 \text{ k}\Omega) \approx 8.01 \text{ M}\Omega$$

Similarly, the downward small-signal resistance of the NMOS cascode current sink is:

$$R_{down} \approx r_{o,MN2} + r_{o,MN4} + g_{m,MN4} \cdot r_{o,MN2} \cdot r_{o,MN4}$$

The transconductance of MN4 is $g_{m,MN4} = 0.45 \text{ mS}$

Thus:

$$R_{down} \approx 281 \text{ k}\Omega + 281 \text{ k}\Omega + (0.00045)(281 \text{ k}\Omega)(281 \text{ k}\Omega) \approx 36.1 \text{ M}\Omega$$

The amplifier output resistance without the load is then:

$$r_{out} = R_{up} \parallel R_{down} \approx 6.56 \text{ M}\Omega$$

Including the load resistance:

$$R_{eff} = r_{out} \parallel R_L \approx 29.9 \text{ k}\Omega$$

Finally, the voltage gain becomes:

$$A_v \approx -g_{m1} \cdot R_{eff} \approx -(3.17 \text{ mS})(29.9 \text{ k}\Omega) \\ A_v \approx -94.8 \text{ V/V}$$

B. Input Resistance r_{in}

The input signal is applied to the gate of MP1. Since the MOSFET gate draws negligible current, the input resistance is approximately infinite: $r_{in} \approx \infty$

C. Output Resistance r_{out}

The small-signal output resistance of the amplifier, excluding the load resistor, is: $r_{out} = R_{up} \parallel R_{down} \approx 6.56 \text{ M}\Omega$

V. CIRCUIT SIMULATION AND PARAMETER TUNING

The circuit designed in Part III was implemented in LTspice using the GF180MCU transistor models. A DC operating point simulation (.op) was performed to verify the bias currents and the output voltage. During simulation, small adjustments were required to achieve the desired operating point. The final parameters used in the simulation are:

Parameter	Initial Value	Final Value
W_{MP1}	2820 μm	2820 μm
W_{MP2}	280 μm	280 μm
W_{MN1}	42 μm	42 μm
W_{MN2}	8.4 μm	8.4 μm
W_{MN3}	42 μm	42 μm
W_{MN4}	8.4 μm	8.4 μm
R_{REF}	7.56 k Ω	10.6 k Ω
V_{BIAS2}	1.65 V	1.65 V
V_{IN}	2.64 V	2.6385 V

The resulting DC operating point from simulation is:

$V_{OUT} \approx 1.66 \text{ V}$ and $I_{REF} \approx 180 \mu\text{A}$ and $I_{BIAS1} \approx 96.5 \mu\text{A}$ and $I_{BIAS2} \approx 41.2 \mu\text{A}$

The simulated results are close to the target design values, and the circuit satisfies the required DC operating conditions.

VI. AC SIMULATIONS

Using the final parameters from Part V, the circuit was simulated in LTspice using AC analysis to obtain the voltage gain and bandwidth. The simulated results are reported in the table below.

Parameter	Calculated	Simulated
Gain $\frac{v_{out}}{v_{in}}$	-94.8 V/V	-12.2 V/V
Input resistance r_{in}	$\approx \infty$	$\approx \infty$
Output resistance r_{out}	6.56 M Ω	$\approx 3 \text{ k}\Omega$
Bandwidth	X	13.8 MHz

VII. COMPARISON BETWEEN CALCULATED AND SIMULATED RESULTS

The simulated results differ from the calculated values because the hand analysis uses simplified small-signal models that neglect non-ideal effects. The LTspice GF180 models include parasitic capacitances, body effect, and channel-length modulation, which affect the gain, output resistance, and bandwidth.